PAPER ID-311035

Subject Code: BME072

Roll No: | | | | | | | | | | | | | |

BTECH
(SEM VII) THEORY EXAMINATION 2025-26
POWER PLANT ENGINEERING

TIME: 3 HRS **M.MARKS: 70**

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A

1. Attempt *all* questions in brief. **02 x 7 = 14**

Q no.	Question	CO	Level
a.	What is meant by carbon credit in power generation systems?	1	K1
b.	What is the function of a condenser in a thermal power plant?	1	K1
c.	Define specific speed of a hydraulic turbine.	2	K1
d.	What is a photovoltaic (PV) cell?	3	K1
e.	Define coefficient of performance of a wind mill rotor.	4	K1
f.	What is the function of an exciter in an electrical generator?	5	K1
g.	Why is earthing necessary in power systems?	5	K1

SECTION B

2. Attempt any *three* of the following: **07 x 3 = 21**

a.	A reheat Rankine cycle operates with steam entering the turbine at 12 MPa, 540 °C and reheated to 540 °C at 3 MPa. Condenser pressure is 0.1 MPa. Calculate the thermal efficiency of the cycle.	1	K3
b.	Describe the control auxiliary systems used for efficient operation.	2	K4
c.	Describe the construction and working of a Pressurized Water Reactor (PWR) with suitable diagram.	3	K4
d.	Explain the working principle of OTEC systems with suitable diagram.	4	K2
e.	Explain the construction and operation of circuit breakers and switchgear.	5	K2

SECTION C

3. Attempt any *one* part of the following: **07 x 1 = 07**

a.	A steam power plant using a simple Rankine cycle produces 100 MW with a cycle efficiency of 35%. Calculate: (a) Heat supplied per second (b) Mass flow rate of steam if heat added per kg is 3000 kJ/kg	1	K4
b.	Explain the working of a condenser and cooling tower in a thermal power plant.	1	K2

4. Attempt any *one* part of the following: **07 x 1 = 07**

a.	Describe the phenomenon of water hammer in penstocks. Explain how surge tanks help in reducing water hammer effects.	2	K4
b.	Describe the combined cycle power plants (CCPP) with suitable diagram.	2	K4

5. Attempt any *one* part of the following: **07 x 1 = 07**

a.	Explain the CANDU heavy water reactor with suitable diagram.	3	K2
b.	Discuss reactor control methods, and radioactive waste disposal techniques in nuclear power plants.	3	K4

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6. Attempt any *one* part of the following: 07 x 1 = 07

a.	Explain the components of a wind energy conversion system with neat sketches.	4	K2
b.	Explain geothermal energy conversion systems with suitable diagram.	4	K2

7. Attempt any *one* part of the following: 07 x 1 = 07

a.	Explain the economics of power generation, including cost of electrical energy, depreciation and energy tariffs.	5	K2
b.	Discuss the environmental aspects of power stations with reference to pollutants, health effects, and pollution control methods.	5	K4

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